Do existing gender and skin tone biases in the fashion and cosmetics industries create unfair outcomes in artificial intelligence for beauty?

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(Dated: May 6, 2025)

The cosmetics and fashion industry is historically subject to gender and skin tone bias. We use exploratory data analysis and statistical tests to examine this bias. Our results indicate that the impact of gender and skin tone on the number of awards earned by a model are statistically significant ($p < 0.05$). We then conduct a qualitative analysis of bias in AI-generated images of fashion models, finding that 52% of the images feature light-skinned models. Finally, we train a clustering algorithm to recognize skin tones and compare it with an industry-standard shade-matching algorithm, revealing that lighting conditions and contrast heavily influence the outcomes.

Keywords: Artificial Intelligence, Fashion, Cosmetics, Fairness, Bias, Skin Tone, Gender

I. INTRODUCTION

The rise of social media has resulted in a watershed moment for the democratization and accessibility of beauty and fashion. As the target markets of these industries have expanded to include women of color, queer people, and men, there has been a growing push for inclusivity in beauty.

Even still, Eurocentrism persists. Features such as pale skin, small noses, and thin bodies are valued over others. Foundation shade ranges reflect a representation gap between light and dark skin tones. Following the release of Fenty Beauty's inclusive foundation line in 2017, this gap has gradually begun to shrink, but bias in cosmetics and fashion still remains. For example, out of 339 Vogue covers published from January 2000 to February 2025, almost 80% featured only white celebrities.

Applications of artificial intelligence (AI) in the beauty industry are varied. Morphe Cosmetics' Lightform foundation shade finder analyzes users' skin tones and generates shade recommendations for the Morphe Extended Hydration Foundation. AI-generated fashion has also found favor with fashion magazines such as *Vogue*, which has promoted *Copy Magazine*, a digital fashion magazine containing AI-generated fashion models.

As AI continues to advance and its widespread adoption by the fashion and cosmetics industries continues, we must examine its potential to reproduce and amplify existing biases within the industry, particularly in terms of skin tone and gender [1].

II. BACKGROUND AND RELATED WORKS

Several research studies have been conducted on the accuracy of skin tone-matching algorithms. One study by Pangelinan et al. compares the efficacy of the Fitzpatrick and Monk skin tone scales for skin tone classification by using colorimeters to measure skin tone in various lighting conditions. Findings included that the Fitzpatrick scale is biased toward lighter skin tones, while the Monk scale is more evenly distributed across light and dark skin tones [2].

Another research study conducted by Narode et al. finds that convolutional neural networks (CNNs) can help improve foundation shade match recommendations by detecting poor lighting [3]. Kadyrova et al. find that the combination of fluorescent lighting and instrument-based color matching produces the most optimal skin tone predictions [4].

Statistical tests in a study conducted by Cynthia M. Frisby find that foundation shade ranges exhibit statistically significant bias in favor of lighter skin tones, offering a greater variety of shades in the light to medium shade range than in the brown to dark shade range [5]. Elizabeth Wissinger's examination of fashion models in New York City suggests that Black fashion models are granted far less job opportunities than white fashion models, with only 2.2% of models featured in the Spring and Summer 2007 collections of 172 different fashion houses being Black [6].

We will first examine these biases by conducting statistical analysis on a dataset of fashion models. We will then investigate whether these biases create unfair outcomes in AI-powered beauty algorithms, specifically in skin tone matching and generative AI.

III. SKIN TONE AND GENDER BIAS IN FASHION

A. Objective

RQ_1: Do skin tone and gender affect the awards and rankings of fashion models?

B. Methodologies

1. Data Collection

We gathered data from Models.com and Fashion Model Directory on 150 fashion models who had won the *Model of the Year* or *Breakout Star* awards from Models.com or were ranked in the Top 50 models for their gender category. Gender identity was confirmed through investigating each model's social media and *Vogue* interviews. To determine the skin tone distribution of our dataset, we gathered *high-quality images* of each model from Models.com and the Fashion Model Directory (FMD). We used the following criteria to determine whether the image was a *high-quality image*: firstly, the photograph must have been taken at a runway show, excluding heavily retouched or edited photos. Secondly, the models themselves must have been *well-lit*, meaning the image was lit from the front and skin tone and facial features were distinguishable. Lastly, if no runway images were available, we used individual discretion to select another campaign image that seemed to represent the model's skin tone well. Using the digital art software Clip Studio Paint, we applied a Gaussian blur with $\sigma = 10$ to each image, then color picked the most homogeneous area of skin. We then manually annotated the models' skin tones based on the Monk Skin Tone scale.

2. Exploratory Data Analysis

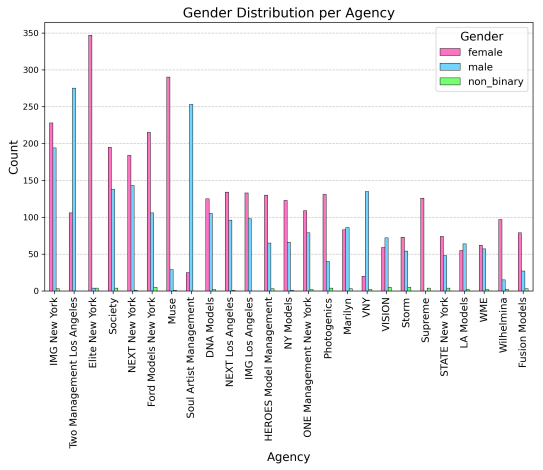


FIG. 1: Snapshot of Distribution of Gender by Agency - Female, Male, Non-Binary

As shown in Figure 1, the vast majority of model agencies have far more female models than male models (with a few exceptions). Non-binary models are poorly represented in the industry, with most agencies employing under 5 non-binary models. The *female* and *male* categories include trans female and trans male models, who also suffer from extremely marginal representation in modeling.

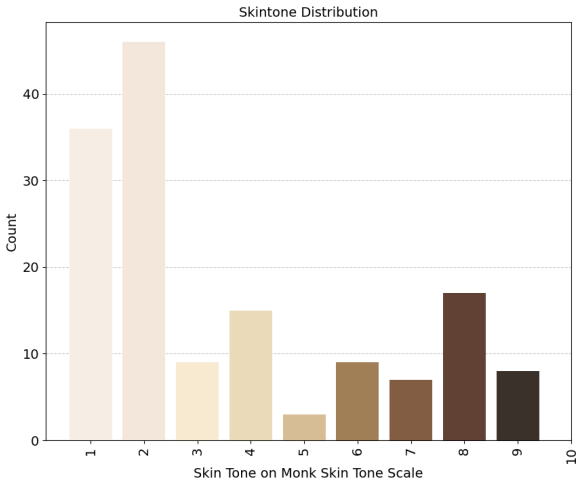


FIG. 2: Distribution of Fashion Model Skin Tones

In Figure 2, which aggregates the skin tones of all models, we can see that lighter skin tones (Monk 1 and 2 specifically) are vastly overrepresented, and there are no models with Monk 10 skin tones in the dataset. While the latter may be attributable to human annotation error, darker skin tones such as Monk 7-9 are still extremely underrepresented, suggesting that there is a systemic bias against darker-skinned fashion models in rankings and awards.

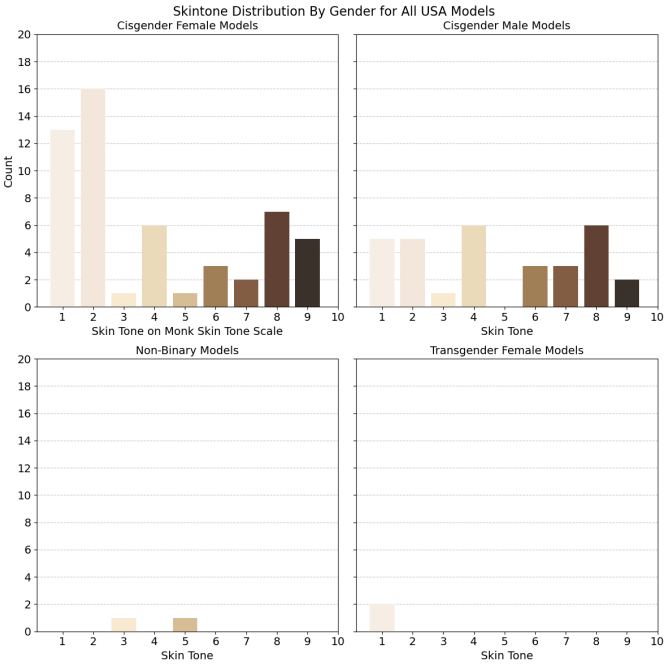


FIG. 3: Skin Tone Distribution by Gender

Lastly, Figure 3 shows the distribution of skin tone separated by gender identity for USA-based models. Gender identities represented in the overall dataset included `F` (Cisgender Female), `M` (Cisgender Male), `NB`

(Non-Binary), `TF` (Transgender Female), and `TM` (Transgender Male). Only `F, M, NB,` and `TF` remained after filtering by country. It appears that there is a far greater bias toward light skin tones (Monk 1-5) among cisgender female models than among cisgender male models, who appear more equally represented across skin tones. Non-binary and transgender female models also have exclusively light skin tones, with absolutely no dark-skinned transgender or non-binary models being represented. This seems to suggest that gender marginalization intensifies the effect of skin tone bias.

3. *Statistical Significance*

H_0 : **We hypothesize that neither gender nor skin tone impact the number of awards that models earn.**

We used ANOVA (Analysis of Variance) to test the above hypothesis. We fit an Ordinary Least Squares linear regression model with `num_achievements` as the response variable and `gender`, `skin_tone`, and their interaction variable `gender:skin_tone` as the predictors.

```
model_awards = smf.ols('num_achievements ~
    C(gender) + C(skin_tone) +
    C(gender):C(skin_tone)',
    data=combined_data).fit()
```

C. Results

The results of our Ordinary Least Squares regression model are shown below.

	Sum Sq	df	F	Pr(>F)
C(gender)	0.0	3.0	0.0	0.0
C(skin_tone)	0.0	6.0	0.0	0.0
C(gender):C(skin_tone)	3.6847437×10^1	18.0	2.599023	0.044809
Residual	2.4416667×10^1	31.0	0.0	0.0

TABLE I: ANOVA on Number of Awards vs. Gender and Skin Tone

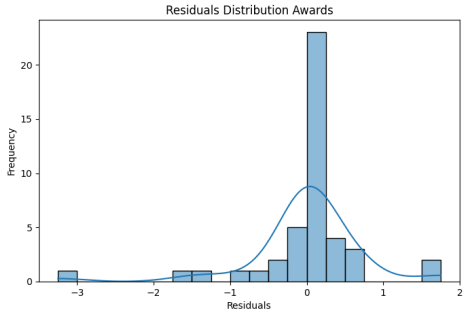


FIG. 4: Residuals of OLS Regression on Awards

The above table and residuals plot indicate that the interaction between gender and skin tone has a highly statistically significant influence ($\Pr(> F) = 0.044809$) on the number of awards earned. This may mean that gender has a differing impact on the number of awards earned by models of different skin tones, or vice versa. The residuals having a $\Pr(> F)$ of 0.0 indicates either a high goodness-of-fit or low variance in the number of awards earned – the latter is possible given that all models earned at least 3 and at most 6 awards.

IV. SKIN TONE AND GENDER BIAS AMONG AI-GENERATED FASHION MODELS

A. Objective

We hypothesize that, given the skin tone and gender biases aforementioned in the fashion modeling industries, similar biases will be reflected in AI-generated fashion campaigns.

RQ_2: Do AI-generated images of fashion models show bias in favor of certain skin tones or gender expressions?

B. Methodologies

1. *Data collection*

We analyzed 81 posts from *Copy* Magazine's official Instagram account. We recorded the date posted, number of likes, caption, a subjective description of the image, the Monk skin tone of the models depicted in the image, the models' assumed ethnicity based on their appearance and their name in the image caption - `white, black, asian,` or `na/ambiguous` - and the slide number of the image in the Instagram post. We did not record assumed ethnicities for real-life models due to the possibility of error – however, given that these AI-generated models were based on text prompts, we hypothesized that these prompts may have included some mention of ethnicity and therefore the variable was relevant.

2. *Results*

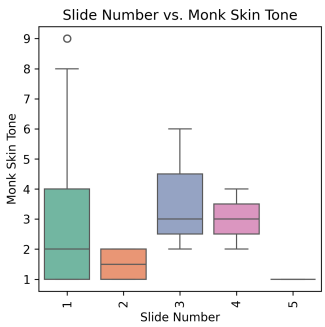


FIG. 5: Instagram Post Slide Number vs. Monk Skin Tone Scale

In Figure 5, we observed that 75% of first-slide images featured models with a skin tone between a Monk 1 and 4. Considering that the first slide of an Instagram post is the "cover" slide that the most viewers will see on their feeds, this indicates minimal effort toward skin tone diversity on the part of *Copy* Magazine's editors.

Slides beyond the first slide nearly uniformly (75-100%) featured models with light skin tones below a Monk 4. Although 78 of the 81 Instagram posts investigated had only one slide, this taken together with the demographics of the first slides does seem to indicate a significant bias against darker skin tones on the part of the magazine's editors.

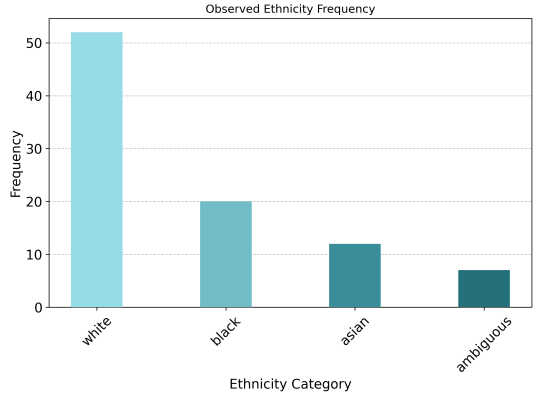


FIG. 6: Frequency of Observed Ethnicities in Copy Magazine posts

We further investigated the frequency of observed ethnicities across *Copy Magazine*'s 81 Instagram posts - 57% of slides featured unambiguously white/Caucasian models.

Conducting a thematic analysis of the recorded image descriptions, 89% featured a lone woman or multiple women. 1 slide featured a lone man. The remainder featured gender-ambiguous models usually wearing feminine clothing, indicating a strong bias in favor of femininity being depicted as the face of fashion (see Figure 1).

V. SKIN TONE BIAS IN COSMETIC SHADE-MATCHING ALGORITHMS

A. Objective

To investigate skin tone bias in the cosmetics industry, we study whether Morphe's Lightform AI-powered Foundation Shade Matcher Quiz ("Lightform") is able to accurately detect various skin tones.

RQ_3: Does the accuracy of the foundation shade recommendations provided by Lightform vary significantly by skin tone, compared to our clustering-based skin tone classifier?

B. Methodologies

1. Data Collection

We used the LFW (Labeled Faces in the Wild) Dataset, sourced from the University of Massachusetts at Amherst (2007), which contains thousands of images of various celebrities' and public figures' faces. To better balance our data, we randomly sampled the images to retrieve exactly one image per subject for a total of 5,750. The images in the dataset are all centered on the subjects' faces and were originally 256×256 pixels in size.

We also collected swatches for each Morphe Lightform Extended Hydration Foundation ("Foundation") shade as our dataset of labels.

2. Preprocessing

We resized all of the images to 128×128 pixels to allow for more efficient training, and normalized the images by dividing the pixels by 255. For each image, we ran the `extract_skin()` function to obtain the average RGB value of a 20×20 square in the center of the image. The image paths and associated skin tone values were then stored in a CSV file.

3. Model

We clustered the images by the calculated skin tone RGB value and extracted the cluster centers, as well as 4 *representative images* per cluster whose skin tones were closest to the cluster center, totaling 40 representative images overall.

We recorded the RGB values of the 36 possible label swatches that Lightform could generate, corresponding to the 36 Foundation shades. We calculated the Euclidean distance between each representative image and each Foundation swatch: $\sqrt{(R_{rep} - R_{swatch})^2 + (G_{rep} - G_{swatch})^2 + (B_{rep} - B_{swatch})^2}$. The 5 Foundation swatches with the minimum distance to each representative image were recorded, for a total of 200 shade matches across the 40 representatives.

The final step of the pipeline was scanning each of the 40 representative images manually with Lightform to determine its calculated shade matches. We used Lightform with a smartphone camera to scan the images, which were displayed on a full-brightness computer screen in natural lighting.

4. Results

The below table compares Equality of Opportunity (**EO**) and Disparate Impacts (**DI**) between different groups of skin tone categories for both models. Too many comparisons were possible between the 10 overall skin tone categories, so we condensed them down to

Metric Type	Skin Tones	Lightform	Clusters
EO	Light/Medium	0.233	0.033
	Medium/Dark	0.200	0.200
	Light/Dark	0.033	0.167
DI	Medium/Light	0.000	0.000
	Dark/Light	0.857	6.000

TABLE II: Fairness Metrics Comparison

three: *Light* covers the skin tones under Morphe's *Fair Light, Light,* and *Light Medium* categories; *Medium* covers Morphe's *Medium, Tan,* and *Deep Tan*; and *Dark* covers Morphe's *Rich, Deep Rich, Deep,* and *Deepest.*

Examining **EO** in particular, our model struggles to distinguish between Medium and Dark skin tones, as well as Light and Dark skin tones. This is likely due to the poor lighting conditions of many images in the LFW dataset causing light skin tones to be misclassified as dark.

In contrast, Lightform struggles to distinguish between Light and Medium skin tones, as well as between Medium and Dark skin tones. However, it is able to distinguish between Light and Dark skin tones fairly well. This suggests that Lightform is unable to pick up on subtler differences between skin tones.

Moving to **DI**, there appears to be no disparity between classifications of Medium and Light skin tones for either model. However, upon further investigation, we found that there were 0 Medium skin tones that were classified accurately by either model. Thus, there is actually quite a large disparity between the correct classification of Light and Medium skin tone groups, favoring Light skin tones.

Lightform classifies Dark skin tones accurately at around 86% of the rate at which it classifies Light skin tones accurately. Our clustering model, on the other hand, shows a *600%* gap favoring Dark tones. This is likely because darker skin tones reflect less light and therefore vary less under different lighting conditions.

The below figures further illustrate the disparity between the two models.



FIG. 7: Shade Recommendations Provided for Alberto Ruiz-Gallardón, former Spanish Minister of Justice (left: Lightform, right: Clustering Model)

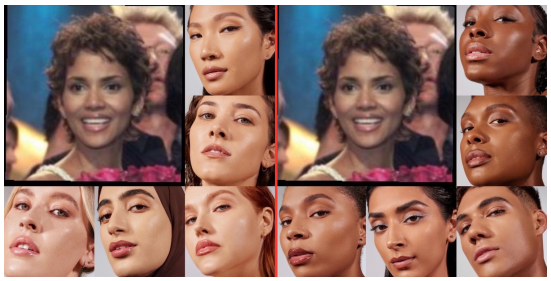


FIG. 8: Shade Recommendations Provided for Halle Berry, American actress (left: Lightform, right: Clustering Model)

VI. CONCLUSION

Our preliminary exploratory data analysis on the fashion model dataset indicated the potential presence of gender bias, skin tone bias, and intersectional bias incorporating both factors. In particular, the distribution of skin tone representation by gender category depicted in Figure 3 suggests that cisgender female, nonbinary, and transgender female models suffer from significant skin tone bias, while cisgender male models are fairly equally represented across skin tones. This could be due to a broader societal association between darker skin tone and masculinity, as outlined in [7].

This intersectional bias was confirmed in our investigation of the number of achievements earned by top fashion models in Table I. While gender and skin tone individually were not effective predictors of the variance in the outcome variable, the interaction between gender and skin tone was a statistically significant predictor.

Finally, our skin tone analysis showed that lighting conditions and contrast in skin tone heavily influence foundation shade recommendations. Fairness measures indicate that Lightform was able to accurately match Light skin tones at the expense of Dark tones, and vice versa for our clustering algorithm. Medium skin tones were poorly classified by both, indicating a need for further investigation in this area.

Our usage of generative AI in the creation of this report was limited to some autocompletion by GitHub Copilot for development productivity and occasional use of ChatGPT to properly format LaTeX tables and figures.

DATA AVAILABILITY

Data for the Fashion Models is available at: `https://www.fashionmodeldirectory.com/` and `https://models.com/`.

Data for AI-generated fashion models is available at: `https://www.instagram.com/magazinecopy/`

Data for Cosmetics Industry Analysis is available at: `https://www.kaggle.com/datasets/jessicali9530/`

lfw-dataset

CODE AVAILABILITY

Code for the exploratory data analysis portion of this paper is available at https://github.com/starryjay/Fairness-In-AI-Fashion.

Code for the skin tone-matching and clustering portion of this paper is available at https://github.com/starryjay/Fairness-In-AI-Skintones.

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